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Bone repositioning guide system and procedure

Abstract

An improved surgical system and procedure for correcting a deformity between first and second bones using an alignment guide based on a correction factor. The correction factor can be based on a virtual model of the first and second bones in a deformed configuration and a corrected configuration. In the virtual corrected configuration, first and second virtual axes can be fixed in the respective first and second bones. When reverted to the virtual deformed configuration, the orientation of the first and second axes can be used to determine the correction factor. The alignment guide is used to insert one or more k-wires into each of the first and second bones in a deformed configuration. A correction guide is passed along the k-wires to rotate and/or translate the first bone relative to the second bone into the corrected configuration.

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Background/Summary

CROSS REFERENCE (1) This application is the national phase application of PCT Patent Application No. PCT/US2020/043525 filed Jul. 24, 2020, which claims the benefit of U.S. Patent Application No. 62/879,340 filed Jul. 26, 2019 the entirety of which is hereby incorporated by reference.

FIELD
(1) The present invention generally relates to surgical systems and procedures for correcting alignment between two bones and a joint and particularly relates to surgical systems and procedures for correcting a bunion in a patient's foot.
BACKGROUND
(2) Bone misalignment and/or deformation can be a source of discomfort and reduced mobility in patients, particularly in a patient's feet. One particularly common foot disorder is a bunion. Bunions are a progressive disorder, typically beginning with a leaning of the great toe. The leaning of the great toe may gradually change an angle of the bones and produce a characteristic bump on the medial side of the metatarsal near the joint of the metatarsal with the proximal phalanx. Specifically, the bunion is the prominence made of bone and at times an inflamed bursa. *Hallux valgus* is the condition in which the great toe deviates from the normal position toward the direction of the second toe. Accordingly, the present invention is directed to surgical systems and procedures for correction of bunions, *Hallux valgus*, and for bone realignments more generally.
SUMMARY
(3) The foregoing summary is illustrative only and is not intended to be limiting. Other aspects, features, and advantages of the systems, devices, and methods and/or other subject matter described in this application will become apparent in the teachings set forth below. The summary is provided to introduce a selection of some of the concepts of this disclosure. The summary is not intended to identify key or essential features of any subject matter described herein.
(4) According to one aspect of the disclosure, a method for correcting alignment between a first bone and a second bone by fusing a joint between the first bone and the second bone includes providing a first guide. The first guide includes a first end portion with a first cannula aligned along a first axis. A second end portion has a second cannula aligned along a second axis. The first axis is non-parallel with the second axis. The first axis is configured to intersect the first bone and the second axis is configured to intersect the second bone when the first and second bones are in a deformed configuration. A first k-wire is inserted through the first cannula and into the first bone. A second k-wire is inserted through the second cannula and into the second bone. The first guide is removed from the first and second k-wires. A second guide includes a first end portion with a first cannula. A second end portion has a second cannula. The first cannula is parallel with the second cannula. The second guide slides over the first and second k-wires. The first k-wire is received within the first cannula of the second guide and the second k-wire is received within the second cannula of the second guide. The second guide acts on the first and second k-wires to re-align the first and second bones into a corrected configuration.
(5) In another aspect, the method includes fixing the first and second bones in the corrected

configuration and removing the second guide and the first and second k-wires from the first and second bones.
(6) In another aspect, the method includes fixing the first and second bones in the corrected configuration includes inserting a first stabilizing wire into the first and second bones.
(7) In another aspect, the method includes attaching a first end of a bone plate with the first bone and a second end of the bone plate with the second bone such that the first and second bones are retained in the corrected configuration.
(8) In another aspect, the method includes inserting a bone plate clip into the first and second bones.
(9) In another aspect, the method includes resecting a first end of the first bone.
(10) In another aspect, the first end portion of the first guide includes a third cannula aligned parallel with the first axis.
(11) In another aspect, the method includes inserting a third k-wire into the first bone through the third cannula and resecting the first end of the first bone includes inserting a first resecting guide over the first and third k-wires to align the first resecting guide with the first end of the first bone.
(12) In another aspect, the method includes resecting a first end of the second bone.
(13) In another aspect, the second end of the first guide includes a fourth cannula aligned parallel with the second axis.
(14) In another aspect, the method includes resecting the first end of the second bone.
(15) In another aspect, the method includes inserting a second resecting guide over the second k-wire and a fourth k-wire inserted in the second bone to align the second resecting guide with the first end of the second bone.
(16) In another aspect, the first bone is a metatarsal, the second bone is a medial cuneiform bone, the deformed configuration of the first and second bones includes a bunion and the corrected configuration of the first and second bones corrects the bunion.
(17) In another aspect, the second guide adjusts an angle of the first bone in three orthogonal planes between the deformed configuration and the corrected configuration.
(18) In another aspect, the second guide adjusts a position of the first bone in three orthogonal planes between the deformed configuration and the corrected configuration.
(19) In another aspect, the method includes centering the first guide between the first bone and the second bone by inserting a centering k-wire through a centering cannula on the first guide.
(20) In another aspect, the method includes removing the first guide from the first and second k-wires includes at least partially disassembling the first guide.
(21) In another aspect, the method includes scanning the first bone and the second bone in the deformed configuration to render a 3D model thereof including a first virtual bone and a second virtual bone in a virtual deformed configuration, adjusting the first virtual bone and the second virtual bone in the 3D model to align the first virtual bone and the second virtual bone in a virtual corrected configuration, fixing a first virtual axis relative to the first virtual bone and fixing a second virtual axis relative to the second virtual bone in the virtual corrected configuration, the first virtual axis is parallel with the second virtual axis, and returning the first and second virtual bones to the virtual deformed configuration, the first and second virtual axes defining a correction factor therebetween in the virtual deformed configuration.
(22) In another aspect, the method includes identifying a virtual resection plane where the first virtual bone and the second virtual bone overlap in the virtual corrected configuration, and fixing the first virtual axis relative to the first virtual bone includes aligning the first virtual axis parallel with the virtual resection plane.
(23) In another aspect, the method includes forming the first guide based on the correction factor.
(24) In another aspect, the correction factor includes a first virtual vector passing through a first virtual point in a virtual coordinate plane and a second virtual vector passing through a second virtual point in the virtual coordinate plane.

(25) In another aspect, forming the first guide includes correlating the virtual coordinate plane with a coordinate plane of the first guide such that the first axis corresponds with the first virtual vector and the first virtual point and the second axis aligned corresponds with the second virtual vector and the second virtual point.
(26) In another aspect, the each of the correction factors includes a position vector and two direction vectors corresponding to the first and second axis of the respective guides within the plurality of guides.
(27) In another aspect, the first guide is selected from a plurality of guides, each of the plurality of guides has a different angle between the first and second axes.
(28) According to another aspect, a method of manufacturing a kit for correcting alignment between a first bone and a second bone includes receiving a correction factor, the correction factor including a first virtual vector passing through a first virtual point in a virtual coordinate plane and a second virtual vector passing through a second virtual point in the virtual coordinate plane.
(29) A first guide is formed based on the correction factor, the first guide including a first end portion having a first cannula aligned along a first axis and a second end portion having a second cannula aligned along a second axis. The first axis corresponds to the first virtual vector and the first virtual point and the second axis corresponds to the second virtual vector and the second virtual point, the first and second axes is non-parallel. The first guide is configured such that a first k-wire inserted through the first cannula intersects the first bone and a second k-wire inserted through the second cannula intersects the second bone in a deformed configuration.
(30) In another aspect, the method includes receiving dimensions of a second guide, the second guide including a first end portion with a first cannula and a second end portion with a second cannula. The first cannula is parallel with the second cannula. The first guide is configured such that when sliding the second guide over the first and second k-wires, the first and second k-wires are received within the respective first and second cannula of the second guide and the second guide re-aligns the first and second bones into a corrected configuration.
(31) In another aspect, the method includes receiving a scan of the first bone and the second bone in the deformed configuration to render a 3D model thereof including a first virtual bone and a second virtual bone in a virtual deformed configuration. The first virtual bone and the second virtual bone are adjusted in the 3D model to align the first virtual bone and the second virtual bone in a virtual corrected configuration. A first virtual axis is fixed relative to the first virtual bone and a second virtual axis is fixed relative to the second virtual bone in the virtual corrected configuration. The first virtual axis is parallel with the second virtual axis. The first and second virtual bones are returned to the virtual deformed configuration along with the first and second virtual axes defining the first and second virtual vectors and the first and second virtual points, respectively, of the correction factor.
(32) In another aspect, the method includes identifying a virtual resection plane where the first virtual bone and the second virtual bone overlap in the virtual corrected configuration and fixing the first virtual axis relative to the first virtual bone includes aligning the first virtual axis parallel with the virtual resection plane.
(33) According to another aspect of the disclosure, a kit for correcting alignment between a first bone and a second bone by fusing a joint between the first bone and the second bone includes a first guide. The first guide includes a first end portion with a first cannula aligned along a first axis and a second end portion with a second cannula aligned along a second axis. The first axis is non-parallel with the second axis. The first guide is configured such that inserting a first k-wire through the first cannula intersects the first bone and inserting a second k-wire through the second cannula intersects the second bone when the first and second bones are in a deformed configuration. A second guide includes a first end portion with a first cannula and a second end portion with a second cannula. The first cannula can be parallel with the second cannula. The second guide is configured such that when the first k-wire is fixed within the first bone and the second k-wire is

fixed within the second bone in the deformed configuration, sliding the second guide over first and second k-wires, with the first and second k-wires is received within the respective first and second cannula of the second guide, re-aligns the first and second bones into a corrected configuration.
(34) In another aspect, a stabilizing wire fixes the first and second bones in the corrected configuration by insertion into the first and second bones.
(35) In another aspect, a bone plate with a first end configured to be attached with the first bone and a second end of the bone plate configured to be attached with the second bone retains the first and second bones in the corrected configuration.
(36) In another aspect, a bone plate clip inserts into the first and second bones in the corrected configuration.
(37) In another aspect, a first resecting guide aligns a resecting tool with a resection location on the first bone.
(38) In another aspect, the first resecting guide includes first and second cannula configured to be advanced over the first k-wire and a third k-wire, the third k-wire is parallel with the first k-wire.
(39) In another aspect, a second resecting guide aligns the resecting tool with a resection location on the second bone.
(40) According to another aspect, a method for correcting alignment between a first bone and a second bone by fusing a joint between the first bone and the second bone includes aligning a first end portion of a first guide with the first bone. The first end portion has a first cannula and a second cannula aligned in a first direction. A first k-wire is inserted through the first cannula and into the first bone and a second k-wire through the second cannula and into the first bone. A first end of the first bone is resected through a slot to form a first resected face. The slot aligns with the first end of the first bone by the first and second k-wires. A third k-wire and a fourth k-wire insert through the first guide into the second bone. A first end of the second bone is resected to form a second resected face. A second guide slides over the first, second, third and fourth k-wires to adjust a positioning of the first and second bones such that the first and second resected faces abut in a corrected configuration. The first and second bones are fixed in the corrected configuration.
(41) In another aspect, the method includes fixing the first and second bones in the corrected configuration by inserting a stabilizing wire into the first and second bones.
(42) In another aspect, the method includes fixing the first and second bones in the corrected configuration by attaching a first end of a bone plate with the first bone and a second end of the bone plate with the second bone such that the first and second bones are retained in the corrected configuration.
(43) In another aspect, the method includes sliding the second guide over the first, second, third and fourth k-wires to translate the first resected face towards the second resected face.
(44) In another aspect, the method includes sliding the second guide over the first, second, third and fourth k-wires to rotate alignment between the first bone and the second bone.
(45) In another aspect, the third k-wire and the fourth k-wire are inserted into the second bone through a second end portion of the first guide including a third cannula and a fourth cannula. The third and fourth cannula are aligned in a second direction.
(46) In another aspect, the first end of the second bone is resected through the slot. The slot is aligned with the first end of the second bone by the third and fourth k-wires.
(47) In another aspect, the slot is on a resection guide including first and second apertures configured to align with the first and second k-wires.
(48) In another aspect, the first bone is a metatarsal, the second bone is a medial cuneiform bone and the corrected configuration of the first and second bones corrects a bunion.
(49) In another aspect, the second guide adjusts an angle of the first bone in three orthogonal planes between a deformed configuration and the corrected configuration.
(50) In another aspect, the method includes removing the first guide from the first and second k-wires after resecting the first end of the second bone to form the second resected face.

(51) In another aspect, the method includes removing the second guide and the first, second, third, and fourth k-wires from the first and second bones after fixing the first and second bones in the corrected configuration.
(52) According to another aspect, a method for correcting alignment between a first bone and a second bone by fusing a joint between the first bone and the second bone includes positioning a cutting guide in a first position proximate to a first end of the first bone, the cutting guide including a cutting slot and a first and second cannula through the cutting guide. The cutting guide in the first position includes a first and second k-wire positioned through the first and second cannula and into the first bone. A first end of the first bone is resected through the cutting slot to form a first resected face. The cutting guide is removed from the first and second k-wires. The cutting guide is positioned in a second position proximate to a first end of the second bone. The cutting guide in the second position includes a third and fourth k-wires positioned through the first and second cannula and into the second bone. A first end of the second bone is resected through the cutting slot to form a second resected face. The cutting guide is removed from the third and fourth k-wires. A second guide slides over the first, second, third and fourth k-wires. The second guide adjusts a positioning of the first and second bones such that the first and second resected faces abut in a corrected configuration. The first and second bones are fixed in the corrected configuration.
(53) In another aspect, positioning a first end portion of a first guide with the first bone, the first end portion having a third cannula and a fourth cannula, the third and fourth cannula aligned in a first direction and inserting the first k-wire through the third cannula and into the first bone and the second k-wire through the fourth cannula and into the first bone.
(54) In another aspect, positioning a second end portion of the first guide with the second bone, the second end portion having a fifth cannula and a sixth cannula, the fifth and sixth cannula aligned in a second direction and inserting the third k-wire through the fifth cannula and into the second bone and the fourth k-wire through the sixth cannula and into the second bone.
(55) In another aspect, fixing the first and second bones in the corrected configuration includes inserting a stabilizing wire into the first and second bones.
(56) In another aspect, fixing the first and second bones in the corrected configuration includes attaching a first end of a bone plate with the first bone and a second end of the bone plate with the second bone such that the first and second bones are retained in the corrected configuration.
(57) In another aspect, sliding the second guide over the first, second, third and fourth k-wires translates the first resected face towards the second resected face.
(58) In another aspect, sliding the second guide over the first, second, third and fourth k-wires rotates the first bone relative to the second bone to adjust an alignment therebetween.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Various examples are depicted in the accompanying drawings for illustrative purposes, and should in no way be interpreted as limiting the scope of the examples. Various features of different disclosed examples can be combined to form additional examples, which are part of this disclosure.
(2) FIG. **1** shows a top view of a patient's foot in a deformed configuration;
(3) FIG. **2**A shows a front perspective view of the alignment guide;
(4) FIG. **2**B shows a rear perspective view of the alignment guide;
(5) FIG. **3**A shows a front view of the alignment guide.
(6) FIG. **3**B shows a section view taken along the line **15**B-**15**B in FIG. **15**A;
(7) FIG. **4** shows an exploded view of the alignment guide;
(8) FIG. **5** shows an angle between cannula of the alignment guide;
(9) FIG. **6** shows a second angle between cannula of the alignment guide;

(10) FIG. **7** shows a third angle between cannula of the alignment guide;
(11) FIG. **8** shows an alignment guide aligned with a medial cuneiform bone and a metatarsal bone in the patient's foot;
(12) FIG. **9** shows insertion of a plurality of k-wires into the medial cuneiform bone and the metatarsal bone through the alignment guide;
(13) FIG. **10** shows a partial disassembly of the alignment guide;
(14) FIG. **11**A shows a perspective view of a resection guide;
(15) FIG. **11**B shows a front view of the resection guide;
(16) FIGS. **12**A-B shows the alignment guide removed and the installation of a first resection guide;
(17) FIG. **13** shows the installation of a second resection guide;
(18) FIG. **14**A shows a perspective view of the correction guide;
(19) FIG. **14**B shows a top view of the correction guide;
(20) FIGS. **15**A-B shows a correction guide assembled over the plurality of k-wires to align the medial cuneiform bone and the metatarsal bone of the patient's foot into a corrected configuration;
(21) FIG. **16** shows the insertion of first and second fixing k-wires into the medial cuneiform bone and the metatarsal bone.
(22) FIG. **17** shows the patient's foot with the plurality of k-wires removed;
(23) FIG. **18** shows an exploded view of a bone plate assembly aligned with the medial cuneiform bone and the metatarsal bone in the corrected configuration;
(24) FIG. **19** shows a top view of a bone plate;
(25) FIG. **20** shows a side view of the bone plate;
(26) FIG. **21** shows the bone plate assembly assembled with the medial cuneiform bone and metatarsal bone in the corrected configuration;
(27) FIG. **22** shows a side view of the patient's foot in the corrected configuration;
(28) FIG. **23** shows a method of calculating a correction factor using a virtual model;
(29) FIG. **24**A shows the virtual model in a virtual deformed configuration;
(30) FIG. **24**B shows the virtual model adjusted into a virtual corrected configuration;
(31) FIG. **24**C shows fixing two virtual axes in a first virtual bone and a second virtual bone, respectively, in the virtual corrected configuration;
(32) FIG. **24**D shows the virtual model returned to the virtual deformed configuration with the resultant orientation of the two virtual axes defining a correction factor for the virtual model;
(33) FIG. **25** shows a method of manufacturing an adjustment guide based on the correction factor;
(34) FIG. **26**A shows a side view of another implementation of an alignment guide;
(35) FIG. **26**B shows a top view of the alignment guide of FIG. **26**A;
(36) FIG. **27** shows an exploded view of the alignment guide of FIG. **26**A;
(37) FIG. **28**A shows a perspective view of another implementation of a resection guide;
(38) FIG. **28**B shows a front view of the resection guide of FIG. **28**A;
(39) FIG. **29** shows alignment of the alignment guide of FIG. **26**A with a patient's foot.
(40) FIG. **30** shows insertion of a plurality of k-wires into a medial cuneiform bone and a metatarsal bone through the alignment guide;
(41) FIG. **31** shows a partial disassembly of the alignment guide;
(42) FIG. **32** shows the alignment guide removed and the installation of the resection guide of FIG. **28**A;
(43) FIG. **33** shows a correction guide assembled over the plurality of k-wires to align the medial cuneiform bone and the metatarsal bone of the patient's foot into a corrected configuration and the insertion of a fixing k-wire.
(44) FIG. **34** shows a bone plate assembly assembled with the medial cuneiform bone and metatarsal bone in the corrected configuration.

DETAILED DESCRIPTION

(45) Overview
(46) Bunion correction or repair is a common surgery with over 100,000 surgeries performed annually in the US. Many surgical procedures for bunion repair are invasive and painful, requiring an incision of several inches and a long period of convalescence, of up to 10-12 weeks. Minimally invasive surgery has been performed in orthopedics for decades. One common procedure is known as a Lapidus bunionectomy. In a Lapidus bunionectomy, the bunion is corrected at the great toe by adjusting alignment at the first tarsometatarsal joint. The metatarsal can also be stabilized using bone screws and/or a plate to facilitate fusion between the metatarsal and the medial cuneiform bone.
(47) However, existing Lapidus bunionectomy procedures have various drawbacks and risks. These drawbacks include requiring more than minimally invasive surgery, the use of a realignment apparatus that exhibits little control over rotation and relative angles of the metatarsal bone, procedures that rely on in-surgery trial-and-error to identify the best alignment of the patient's foot bones and in-surgery judgment to identify locations for performing resections, lack of customization to account for individual patient foot conditions, and/or a lack of usable guides for performing pre-planned resections of the foot bones. Various aspects of the bone repositioning systems and procedures described herein overcome and improve upon these existing procedures, leading to better patient outcomes.
(48) The various features and advantages of the systems, devices, and methods for bone repositioning described herein will become more fully apparent from the following description of the examples illustrated in the figures. These examples are intended to illustrate the principles of this disclosure, and this disclosure should not be limited to merely the illustrated examples. The features of the illustrated examples can be modified, combined, removed, and/or substituted as will be apparent to those of ordinary skill in the art upon consideration of the principles disclosed herein.
(49) Deformation Correction Procedures
(50) FIG. **1** shows a skeletal view of a patient's foot **100** having one or more bones in a deformed configuration **102**. The deformed configuration **102** can be a bunion, as illustrated. The deformed configuration **102** can be a misalignment between a metatarsal **108** and a phalanx **112** of the patient's great toe. The metatarsal **108** can be at an angle with respect to the phalanx at **112**. A high degree of misalignment between the metatarsal **108** and the phalanx **112** can lead to severe pain and rubbing and discomfort and other problems in the patient's foot **100**. Accordingly, it can be beneficial to correct the alignment between the metatarsal **108** and the phalanx **112** of the great toe.
(51) The patient's foot **100** can further include a medial cuneiform bone **104**. The medial cuneiform bone **104** can be connected with a proximal end of the metatarsal **108** (e.g., by one or more ligaments). FIGS. **1-22** illustrate systems and methods of correcting alignment between the medial cuneiform bone **104** and the metatarsal **108**. In turn, proper alignment between the medial cuneiform bone **104** and the metatarsal **108** can correct alignment between the metatarsal **108** and the phalanx **112**. Accordingly, the deformed configuration **102** of the patient's foot **100** can be corrected. The present disclosure relates to systems and methods for correcting the deformed configuration **102**. Moreover, the systems and methods described herein can be used more generally for correcting alignment between any two bones a patient's body.
(52) As shown in FIGS. **2**A-**4**, the system for correcting alignment in the patient's foot **100** can include an alignment guide **200**. The alignment guide **200** can be formed of a rigid material. The alignment guide **200** can include a first end portion **204**. The first end portion **204** can include one or more apertures **210***a*, **212***a*. Although two apertures are described and illustrated, more or fewer apertures can be included on the first end portion **204**. The apertures **210***a*, **212***a* can include internal threads **221**, **222**, respectively. The apertures **210***a*, **212***a* can be chamfered on one or both sides of the alignment guide **200**. The apertures **210***a*, **212***a* can extend all the way through the alignment guide **200**. The apertures **210***a*, **212***a* can be aligned along respective axes **230**, **232**. The

axes **230**, **232** can be parallel. Alternatively, the axes **230**, **232** can be converging. The axes **230**, **232** can be spaced apart a distance **204***a*. The distance **204***a* can be based on a length of the medial cuneiform bone **104**.

(53) As shown in FIG. **4**, the alignment guide **200** can include one or more removable tubes **240**, **242**. The removable tubes **240** can include a first end **240***a* and a second end **240***b*. The first end **240***a* can be received within the aperture **210***a*. The removable tube **240** can include a threaded portion **244**. The threaded portion **244** can engage with the internal threads **221** of the aperture **210***a*. The removable tubes **242** can include a first end **242***a* and a second end **242***b*. The first end **242***a* can be received within the aperture **212***a*. The removable tube **240** can include a threaded portion **246**. The threaded portion **246** can engage with the internal threads **222** of the aperture **212***a*.

(54) The removable tube **240** can define a cannula **210**. When installed within the aperture **210***a*, the cannula **210** can be aligned along the axis **230** of the aperture **210***a*. The removable tube **242** can define a cannula **212**. When installed within the aperture **212***a*, the cannula **212** can be aligned along the axis **232** of the aperture **212***a*. The cannula **210**, **212** can define different diameters therethrough. The cannula **212** can have a greater diameter than the cannula **210** (or vice-versa). In other implementations, the cannula **210**, **212** can define the same different diameters therethrough. In other implementations, the cannula **210**, **212** can define varying diameters therethrough.

(55) The alignment guide **200** can include a second end portion **208**. The second end position **208** can include one or more cannula **214**, **216**. The cannula **214**, **216** can be defined through a body of the alignment guide **200** and/or through respect extensions **219**, **218** thereof. Although two cannula are described and illustrated, more or fewer cannula can be included on the second end portion **208**. Moreover, the second end portion **208** can include removable inserts or removable portions (e.g., removable tubes) around the cannula **214**, **216**.

(56) The cannula **214**, **216** can extend all the way through the alignment guide **200** (e.g., including the extensions **218**, **219**). The cannula **214**, **216** can define different diameters therethrough. The cannula **214** can have a greater diameter than the cannula **216** (or vice-versa). In other implementations, the cannula **214**, **216** can define the same different diameters therethrough. In other implementations, the cannula **214**, **216** can define varying diameters therethrough.

(57) The cannula **214**, **216** can be aligned along respective parallel axes **234**, **236**. The axes **234**, **236** can be spaced apart a distance **208***a*. The distance **204***a* can be based on a length of the metatarsal bone **108**.

(58) FIGS. **5-7** shows the assembled alignment guide **200**. The first end portion **204** can define a positioning and orientation of a first set of cannula (e.g., cannula **210**, **212**). The second end portion **208** can define a positioning and orientation of a second set of cannula (e.g., cannula **214**, **216**). The first set of cannula and the second set of cannula can be offset from each other and/or angled with respect to each other.

(59) FIG. **5** shows an angle α between the axis **230** of the cannula **210** and the axis **236** of the cannula **216**. The angle α defines the relative orientation angle between the first set of cannula on the first end **204** and the second set of cannula on the second end **208**. The angle α can be defined in an z-x plane in a Cartesian coordinate system (having x, y, and z axes). The cannula **210** can include a point A. Alternatively, the point A can be any fixed position along the cannula **210**. The point A can have an x, y, and z coordinate location in a Cartesian coordinate system (having x, y, and z axes). The cannula **216** can include a point B. Alternatively, the point B can be any fixed position along the cannula **216**. The point B can have an x, y, and z coordinate location in the Cartesian coordinate system. The points A and B can define a relative position of the axes **230**, **236** in the Cartesian coordinate system.

(60) FIG. **6** shows an angle β between the axis **230** of the cannula **210** and the axis **236** of the cannula **216**. The angle β defines the relative orientation angle between the first set of cannula on the first end **204** and the second set of cannula on the second end **208** in a y-x plane. FIG. **7** shows

an angle γ between the axis **230** of the cannula **210** and the axis **236** of the cannula **216**. The angle γ defines the relative orientation angle between the first set of cannula on the first end **204** and the second set of cannula on the second end **208** in a y-z plane.
(61) Together, the relative positions of the points A and B and at least two of the relative angles α, β, and γ can define the axis of the cannula on the alignment guide **200**. Using the proper selection of the relative angles α, β, and/or γ, and/or the relative positions of the points A, B, the alignment guide **200** can be used to correctly align the bones in the patient's foot **100**, as described further below.
(62) As shown in FIG. **8**, the alignment guide **200** can be aligned with the patient's foot **100**. The first end portion **204** can be generally aligned with the medial cuneiform bone **104**. The second end portion **208** can be generally aligned with the metatarsal **108**. As shown in FIG. **9**, a plurality of k-wire **300** can be extended through respective cannula of the alignment guide **200**. The k-wires can extend through the cannula and into the respective medial cuneiform **104** and the metatarsal bone **108**. A first k-wire **310** can be inserted through the cannula **210** and into the medial cuneiform **104**. The first k-wire **310** can be inserted at an insertion point **320** on the medial cuneiform bone **104**. A second k-wire **312** can be inserted through the cannula **212**. The second k-wire **312** can be inserted through the medial cuneiform bone **104** at an insertion point **322**. A third k-wire **314** can be inserted through the cannula **214**. The third k-wire **314** can intersect and be inserted into the metatarsal **108** at an insertion point **324**. A fourth k-wire **316** can be inserted through the fourth cannula **216**. The fourth k-wire **316** can be inserted into the metatarsal **108** at an insertion point **326**.
(63) The first and second k-wires **310**, **312** can be parallel with each other, based on the parallel cannula **210**, **212**. The third and fourth k-wires **314**, **316** can be parallel with each other, based on the cannula **214**, **216**. One or more of the insertion points **320**, **322**, **324**, **326** (e.g., at least one on each bone **104**, **108**) can be in predetermined locations on the patient's foot. The lengths of the extensions **218**, **219** and/or the tubes **240**, **242** can provide greater stability to the k-wires **300** that are received therein. Diameters of the k-wires **300** can be sized according the diameters of the respective cannula of the alignment guide **200** to ensure accurate insertion at angles into the bone **104**, **108**. Moreover, the k-wires **300** can be matched to the correct cannula based on different diameter sizes.
(64) FIG. **10** shows removal of the tubes **240**, **242** from the first end **204** of the alignment guide **200**. The first and second tubes **240**, **242** are removed from the first end portion **204** to enable the alignment guide **200** to be removed from the plurality of k-wires **300** inserted within the medial cuneiform bone and metatarsal bone **108**. In certain circumstances, without a removable or otherwise deconstructable element, it can be difficult for a user to remove the alignment guide **200** from the plurality of k-wires **300** because of the misalignment between the first and second ends **204**, **208**.
(65) As shown in FIGS. **11**A-B, the system for correcting alignment in the patient's foot **100** can include a resection guide **404**. The resection guide **404** can align a resecting tool (not shown), such as a saw, a broach or the like, with an end of the medial cuneiform bone **104** and/or an end of the metatarsal bone **108**, respectively.
(66) The resection guide **404** can include a cannulated portion **411**. The cannulated portion **411** can include one or more apertures **415**, **417**. The resection guide **404** can include a plane portion **409**. The plane portion can include a slot **407** for aligning the resecting tool. The apertures **415**, **417** can interact with one or more k-wires (e.g., k-wires **300**) or pins to align the plane portion **409** with the desired target location for the resection tool.
(67) The plane portion **409** (e.g., a plane defining the slot **407**) can be generally perpendicular with the cannulated portion **411** (e.g., an axis between the apertures **415**, **417**). In other implementations, the plane portion **409** can be angled with respect to the cannulated portion **411**.
(68) The apertures **415**, **417** can extend through the cannulated portion **411**. The apertures **415**, **417**

can be sized to align with the k-wires or pins. The slot **407** can extend through the plane portion **409**. The slot **407** can have a height and thickness sized to accommodate the cutting portion of the resection tool. The slot **407** can have a depth sufficient to maintain alignment of the resection tool with the desired target location.

(69) Depending on a planned corrected configuration of the first cuneiform bone **104** with the metatarsal **108**, it may be necessary to remove material from one or both inner ends of the cuneiform bone **104** and the metatarsal **108**. The angle between the cuneiform bone **104** and the metatarsal **108** can be adjusted in the corrected configuration. The lengths of one or both of the cuneiform bone **104** and the metatarsal **108** can also be adjusted in the corrected configuration. Each of these adjustments can contribute to the correction of the deformity in the patient's foot **100**.

(70) Accordingly, FIGS. **12**A-B show usage of the first resection guide **404** to align a resection tool with a first inner end of the first cuneiform bone **104**. The cannulated portion **411** can be received over the first and second k-wire **310**, **312** on the apertures **415**, **417**, respectively. This can align the plane portion (e.g., the slot **407**) with the end of the first cuneiform bone **104**. A resection plane **104***a* can be cut into the first cuneiform bone **104** using a resecting tool through the slot **407**. The resection plane **104***a* can be aligned with the first and second k-wire **310**, **312**.

(71) FIG. **13** shows usage of a second resection guide **408** to align a resection tool with a first inner end of the metatarsal bone **108**. The second resection guide **408** can include the same components as the resection guide **404** (e.g., a plane portion **409** and a cannulated portion **411**).

(72) The cannulated portion **411** of the second resection guide **408** can be received over the third and fourth k-wires **314**, **316** on apertures **415**, **417**, respectively. The third and fourth k-wires can align the plane portion **409** and a slot **407** with the end of the metatarsal bone **108**. A resection plane **108***a* can be cut into the metatarsal bone **108** using a resecting tool through the slot **407**. The resection plane **108***a* can be aligned with the third and fourth k-wires **314**, **316**. In some implementations, the resection guide **404** can be used to form the resection plane **108***a* instead of the section resection guide **408**.

(73) As shown in FIGS. **14**A-B, the system for correcting alignment in the patient's foot **100** can include a correction guide **500**. The correction guide **500** can align the bones in the patient's foot **100** into a corrected configuration **103**, as shown in FIG. **15**. The correction guide **500** can include a first end portion **504**. The first end portion **504** can include one or more cannula **510**, **512**. The cannula **510**, **512** can extend through the correction guide **500**. The cannula **510**, **512** can correspond to the first and second k-wires **310**, **312**, respectively inserted within the medial cuneiform bone **104**. The cannula **510**, **512** can be aligned along first and second axes **530**, **532**, respectively. The first and second axes **530**, **532** can be parallel.

(74) The correction guide **500** can include a second end portion **508**. The second end portion **508** can include one or more cannula **514**, **516**. The cannula **514**, **516** can extend through the correction guide **500**. The cannula **514**, **516** can correspond to the third and fourth k-wires **314**, **362**, respectively inserted within the metatarsal bone **108**. The cannula **514**, **516** can be aligned along third and fourth axes **534**, **536**, respectively. The third and fourth axes **534**, **536** can be parallel.

(75) The first and second axes **530**, **532** can be parallel with the third and fourth axis **534**, **536**. The first and second axes **530**, **532** can aligned within the same plane as the third and fourth axis **534**, **536**. In other implementations, the first and second axes **530**, **532** can be aligned out of plane with (e.g., offset from) the third and fourth axis **534**, **536**.

(76) The first axis **530** can extend through a point C. The point C can have a position (x, y, z) in the Cartesian coordinate plane. The fourth axis **536** can extend through a point D. The point D can have a position (x, y, z) in the Cartesian coordinate plane. The points C and D can define a relative position of the first and fourth axes **530**, **536** in the Cartesian coordinate system.

(77) As shown in FIG. **15**, the correction guide **500** can be received on the k-wires **300**. The first and second k-wires **310**, **312** can be received within the cannula **510**, **512**, respectively, on the first end **504** of the correction guide **500**. The third and fourth k-wires **314**, **316** can be received within

the cannula **514**, **516**, respectively, on the second end **508** of the correction guide **500**.
(78) Accordingly, the correction guide **500** can align the metatarsal bone **108** relative to the medial cuneiform bone **104** as it is advances on the k-wires **300**. The correction guide can orient the metatarsal bone **108** and the proximal phalanx **112** into the corrected configuration **103**. Re-orientation of the metatarsal **108** relative to the medial cuneiform bone **104** can include rotation and/or translation of the metatarsal **108** in the Cartesian coordinate system (e.g., in three orthogonal planes). The degree of rotation and/or translation of the metatarsal **108** can be determined based on the angles α, β, and/or γ, and/or any differences in the relative positions of the axes between the alignment guide **200** and the correction guide **500** (e.g., any differences in the relative positions defined by points A, B and points C, D).
(79) The corrected configuration **103** can include one or more corrections to the alignment of the bones of the patient's foot **100**. For example, the metatarsal **108** can be generally aligned with the proximal phalanx **112** of the great toe. The corrected configuration **103** can promote healing of the bunion and/or *hallux valgus* deformity. The resected face **104***a* of the medial cuneiform bone **104** can be abutted against the resected face **108***a* of the metatarsal bone **108**. This abutment can promote the union or fusion of the metatarsal **108** with the medial cuneiform bone **104**. Proper abutment can require translation of the metatarsal **108** relative to the medial cuneiform bone **104**.
(80) As shown in FIG. **16**, the medial cuneiform bone **104** can be fixed temporarily or permanently relative to the metatarsal **108** in the corrected configuration **103**. A first fixing k-wire **610** can be inserted into the medial cuneiform bone **104** and the metatarsal bone **108**. The first fixing wire **610** can extend through the resection faces **104***a*, **108***a*. A second fixing k-wire **612** can be inserted through the metatarsal **108** and into the medial cuneiform bone **104**. The second fixing k-wire **612** can be inserted through the resection planes **104***a*, **108***a*. In other implementations, any temporary or permanent fixing means can be used for connecting the medial cuneiform bone **104** with the metatarsal bone **108** in the corrected configuration. For example, the medial cuneiform bone **104** and the metatarsal bone **108** can be screwed together, braced together, adhered together or otherwise connected together on a temporary or permanent basis.
(81) As shown in FIG. **17**, with the medial cuneiform bone **104** and the metatarsal bone **108** fixed in the corrected configuration **103**, the correction guide **500** can be removed from the plurality of the k-wires **300**. The plurality of k-wires **300** can be removed from the medial cuneiform bone **104** and/or the metatarsal bone **108**.
(82) As shown in FIG. **18**, the system for correcting alignment in the patient's foot **100** can include a bone plate assembly **700**. The bone plate assembly **700** attach the medial cuneiform bone **104** and the metatarsal **108**, as shown in FIGS. **21-22**. The bone plate assembly **700** can include a bone plate **710**. The bone plate **710** can include a first end **704** and a second end **708**. The bone plate assembly **700** can include a bone clip **720**. The bone clip **720** can couple between the medial cuneiform bone **104** and the metatarsal bone **108**. The bone clip **720** can include a first prong **724** and second prong **728** connected by a transverse member **726**. The bone plate assembly **700** can include a plurality of fasteners **730** such as bone screws, pins or other fasteners known in the field of orthopedics.
(83) FIGS. **19-20** show further detail of the bone plate **710**. The bone plate **710** can be contoured to fit against the medial cuneiform bone **104** and the metatarsal bone **108**. The bone plate **710** can to be made out of titanium, aluminum, steel, other suitable materials in the orthopedic field.
(84) The first end **704** of the bone plate **710** can have a plurality of apertures **715**, **716**, **717**. The apertures **716**, **717** can be sized to receiver the fasteners **730**. The aperture **715** can be sized to receive the prong **724** of the clip **720**. The second end **708** of the bone plate **710** can have a plurality of apertures **711**, **712**, **713**. The apertures **711**, **712** can be sized to receive the fasteners **730**. The aperture **713** can be sized to receive the prong **728** of the clip **720**. The clip **720** can include a recess **719** for receiving, or at least partially receiving, the transverse member **726** of the clip **720**. This can reduce the overall profile of the assembled bone plate assembly **700**.
(85) FIGS. **21-22** show the bone plate assembly **710** assembled with the patient's foot **100**. The first

end **704** of the bone plate **710** can be attached with the medial cuneiform bone **104** by the fasteners **730**. The fasteners **730** can extend through the apertures **716**, **717** and into the medial cuneiform bone **104**. The second end **708** of the bone plate **710** can be attached with the metatarsal bone **108**. The fasteners **730** can extend through the apertures **711**, **712** and into the metatarsal bone **108**. In some implementations, the fasteners **730** can be received within respective intersection points **320**, **322**, **324**, and/or **326** of the k-wires **300**. Alternatively, the fasteners can form new holes in the bones of the patient's foot.
(86) The clip **720** can span across the joint between the medial cuneiform bone **104** and the metatarsal bone **108**. The first prong **724** can be received within aperture **715** and into the metatarsal bone **108**. The second prong **728** can be received through the aperture **713** and into the medial cuneiform bone **104**. In certain implementations, the prongs **724**, **728** can be received within the respective intersections **322**, **324**. The prongs **724**, **728** can include a plurality of serrated edges for enhanced engagement features for attaching within the bones in the patient's foot **100**.
(87) In certain implementations, different alignment guides **200** can be used depending on the intended fixation means for the medial cuneiform bone **104** with the metatarsal bone **108**. The different alignment guides **200** can include cannula that align the k-wires **300** at different points in the bone to match apertures in the different fixation means.
(88) Virtual Modelling of Correction Factor
(89) FIG. **23** describes a process **800** for designing an alignment guide that customized to a patient's unique anatomy. Although described herein in the context of a patient's foot, the process **800** can be used for other parts of a patient's body. The process **800** is further illustrated in FIGS. **24**A-**24**D. At step **812**, a virtual model **840** of a patient's foot is created. The virtual model **840** can be based on a scan of a patient's foot including a deformity, such as a bunion and/or *Hallux valgus*. The scan used to create or render the virtual model **840** can be based on a CT, PET, X-ray, ultrasound, MRI, or other type of medical imaging scan.
(90) The virtual model **840** can include virtual representations of the bones of the patient's foot. The virtual model **840** can include a virtual deformed configuration **802** of the patient's bones. The virtual model **840** can include a virtual first bone **804** and a virtual second bone **808**. The virtual first bone **804** can correspond to a medial cuneiform bone in the patient's foot and the virtual second bone **808** can correspond to a metatarsal.
(91) The virtual model **840** can be displayed to a user through a graphical user interface (e.g., on a computer). The virtual model **840** can be manipulable by a user. In some implementations, the virtual model **840** can approximate the natural connections (e.g., ligaments, cartilage, and/or muscles) between bones in the patient's foot. Accordingly, movements of one virtual bone can alter the location of connected virtual bones. In other implementations, the virtual bones of the model **840** can be freely moved and manipulated by a user. Accordingly, feasible re-positioning of the bones and resultant movements of connected virtual bones can be approximated based on the skill and knowledge of a user.
(92) At step **814**, a user adjusts the configuration of first and second virtual bones **804**, **808** into a virtual corrected configuration **803**. The virtual corrected configuration **803** can include correction of one or more deformities of the patient's foot. Adjustment into the virtual corrected configuration **803** can include changing relative angles and positions between the first and second virtual bones **804**, **808**. Moreover, the virtual corrected configuration **803** can include one or more overlapping portions of the first and second virtual bones **804**, **808**. One or more virtual resection planes **804***a*, **808***a*, can be identified by a user to remove overlapping portions of the first and second virtual bones **804**, **808** or otherwise adjust the lengths and dimensions thereof.
(93) At step **816**, a first virtual axis **830** is added to intersect the first virtual bone **804**. A second virtual axis **836** is added to intersect the second virtual bone **808**. The first virtual axis **830** is fixed relative to the first virtual bone **804**. The second virtual axis **836** is fixed relative to the second virtual bone **808**. The first and second virtual axes **830**, **836** can be aligned with the virtual model

840 at a location that is easily accessible during surgery of the patient's foot.
(94) The first and second virtual axes **830**, **836** are parallel with each other. Advantageously, the first and second virtual axes **830**, **836** can be aligned with one or more of the virtual resection planes **804***a*, **808***a*. The first virtual axis **830** extends through a point G located in a virtual Cartesian coordinate system. The second virtual axis **836** extends through a point H located in the virtual Cartesian coordinate system.
(95) At step **818**, the first and second virtual bones **804**, **808** are returned to the original deformed configuration **802** of the model **840**. The first and second virtual axes **830**, **836** are rotated to different angles and/or translated relative to each other into the deformed configuration **802** from the corrected configuration **803**. In the deformed configuration **802**, the first and second virtual axes **830**, **836** can be defined as vectors passing through respective point E, F, respectively, within the virtual Cartesian coordinate system.
(96) At step **820**, the relative positions of the first and second virtual axes **830**, **836** in the deformed configuration **802** can be used to define a correction factor for an alignment guide. The relative positions can include relative angles in two or more of the virtual Cartesian coordinate system planes (e.g., z-x, z-y, x-y). The relative angles can correspond to the α, β, and/or γ angles in the alignment guide (e.g., alignment guide **200** or the like). The relative positions of the first and second virtual axes **830**, **836** can be based on the respective point E, F. The points E, F can correspond to the respective points A, B in the alignment guide (e.g., alignment guide **200** or the like). Thus, the dimensions of the virtual model **840** can be used to form the correction factor of an alignment guide for use in surgery on the patient's foot.
(97) Furthermore, the relative positions of the first and second virtual axes **830**, **836** in the corrected configuration **803** can be used to define dimensions of a correction guide. The points G, H can correspond to the respective points C, D in the correction guide (e.g., correction guide **500** or the like). The first and second virtual axes **830**, **836** in the corrected configuration **803** can correspond to the parallel axes of the cannula in the correction guide.
(98) Furthermore, the relative positions of the first and second virtual axes **830**, **836** in the deformed configuration **802** can be used to define dimensions of a resection guide. The dimensions can include an orientation of a slot (e.g., slot **407**) in the resection guide. The slot can be aligned parallel with one or more of the resection planes **804***a*, **808***a*. The resection guide can also include one or more apertures aligned with the first and/or second virtual axes **830**, **836** in the deformed configuration **802**.
(99) As an alternative to creating the model **803**, a user (e.g., a surgeon) could describe the angles (α, β, and/or γ) and/or translations needed to correct the deformities in the patient's foot **100**. This description can be based on a user's knowledge and experience and/or in conjunction with viewing a scan of the patient's foot **100**. The user-provided information can indicate the alignment guide **200** needed during in surgery. For example, a user can be provided a kit with multiple alignment guides and select among a pre-determined set of alignment guides **200** that each correct different, but commonly seen deformations in a patient's foot. In certain implementations, the alignment guide **200** can include multiple sets of cannula that correspond to different correction factors.
(100) Manufacturing of Lapidus System
(101) Process **900** is a method of manufacturing a system for correcting alignment in the patient's foot **100** based on a correction factor. At step **912**, a manufacturer can receive a correction factor. The correction factor can define one or more dimensions of an alignment guide (e.g., alignment guide **200**). The correction factor can be a CAD model, in some implementations. The dimensions can include orientation and positioning of one or more cannula therethrough. For example, the correction factor can be based on the process **800** described above and/or user-provided information. The correction factor can be customized to an individual patient's foot. Alternatively, the correction factor can be one of a standard set of commonly used correction factors.
(102) At step **914**, the manufacturer can form the alignment guide based on the correction factor.

For example, the manufacture can 3D print the alignment guide.
(103) At step **916**, the manufacturer can receive dimensions for creating a correction guide. The dimensions for the correction guide can be based on the process **800** described above or otherwise customized to an individual patient's foot.
(104) At step **918**, the manufacturer can form the correction guide based on the received dimensions. For example, the manufacture can 3D print the correction guide.
(105) Alternative Component Structures
(106) FIGS. **26**A-**26**B illustrate another possible configuration for an adjustment guide **1000**. The adjustment guide **1000** can include the same features and functionalities of the adjustment guide **200** described above, including some of the differences noted below. The adjustment guide **1000** can include a first portion **1004** and a second portion **1008**. The first portion **1004** can be releasably connectable with the second portion **1008**. The first portion **1004** can include a handle **1004***a*. The handle portion **1004***a* can include an aperture therethrough. The handle portion **1004***a* can function to enable a user to easily hold the adjustment guide **1000** in place during use. The first portion **1004** of the adjustment guide **1000** can include one or more cannula **1010**, **1012** extending therethrough. The cannula **1010**, **1012** can extend through the first portion **1004**. The cannula **1010**, **1012** can extend along parallel axes **1020**, **1022**.
(107) The second portion **1008** can include one or more cannula **1014**, **1016**. The cannula **1014**, **1016** can extend through the second portion **1008**. The cannula **1014**, **1016** can extend along parallel axes **1024**, **1026**, respectively. The axes **1020**, **1022** can be nonparallel with the axes **1024**, **1026**.
(108) The first or second portions **1004**, **1008** can include a centering cannula **1009**. The centering cannula **1009** can be used to align the adjustment guide **1000** at the tarsometatarsal joint between the medial cuneiform bone **104** and the metatarsal **108**.
(109) As shown in FIG. **27**, the first portion **1004** can be connectable with the second portion **1008** by an attachment mechanism **1006**. The attachment mechanism **1006** can be a thumbscrew. As a thumbscrew, the attachment mechanism **1006** can include a threaded end **1006***a*. The attachment mechanism **1006** can extend through an aperture **1006***b* in the first portion **1004**. The attachment mechanism **1006** can extend through an aperture **1006***c* in the second portion **1008**. At least one of the apertures **1006***b*, **1006***c* can be internally threaded to couple with the threaded end **1006***a*. Accordingly, the first and second portions **1004**, **1008** can be coupled together by the attachment mechanism **1006**.
(110) The second portion **1008** can include a recess **1008***a*. The first portion **1004** can include a projection portion **1004***b*. The projection portion **1004***b* can be received within the recessed portion **1008***a*. The recess/protrusion arrangement can enhance the stability of the coupling between the first portion **1004** and the second portion **1008**.
(111) FIGS. **28**A-**28**B show another implementation of a resection guide **1100**. The resection guide **1100** can be structured similarly to the resection guide **404** described above, including some of the differences noted herein. The resection guide **1100** can include a first portion **1111**. The first portion **1111** can include one or more apertures **1115**, **1117** extending therethrough. The first portion **1111** can be coupled with a planar portion **1109**. The planar portion **1109** can include a slots **1107** therein. The slot **1107** can be sized to allow a resection tool to extend therethrough for resecting a bone of a patient's body (e.g., the patient's foot **100**). In some implementations, the planar portion **1109** can include a curved shape to allow the slot **1107** to be placed closer to and/or in contact with the patient's body. This can reduce error associated with the process of resecting a bone.
(112) FIG. **29** shows one method of using the adjustment guide **1100** in a procedure for correcting alignment between two bones in a patient's body. The adjustment guide **1000** can be used to correct alignment of a medial cuneiform bone **104** and a metatarsal bone **108** in a patient's foot **100**. The process shown in FIGS. **29-34** is similar to and can include any of the steps and details described above in the process shown in FIGS. **1-22**.

(113) The centering cannula **1009** can align the adjustment guide **1000** at the tarsometatarsal joint between the medial cuneiform bone **104** and the metatarsal **108**. A k-wire (not shown) can extend through the centering cannula **1009** and into the space between the medial cuneiform bone **104** and the metatarsal **108**. The first end **1004** of the adjustment guide **1000** can be generally aligned with the medial cuneiform bone **104**. The second end **1008** of the adjustment guide **1000** can be generally aligned with the metatarsal **108**. As shown further in FIG. **30** a plurality of k-wires **1300** can be inserted through the respective cannula of the adjustment guide **1000** and into the medial cuneiform bone **104** and the metatarsal bone **108**. A first k-wire **1310** can be received within the cannula **1010** and intersect the medial cuneiform bone **104**. A second k-wire **1312** can be received through the cannula **1012**. A third k-wire **1314** can be inserted through the cannula **1014** and into the metatarsal **108**. A fourth k-wire **1316** can extend through the cannula **1016** into the metatarsal **108**. The k-wires **1300** can extend along respective axes of the cannula of the adjustment guide **1000**. Accordingly, the alignment guide can define the intersection angles of the k-wires **1300**.
(114) As shown in FIG. **31**, the first portion **1004** of the adjustment guide **1000** can be removed from the second portion **1008**. The attachment mechanism **1006** can be removed from between the first portion **1004** and the second portion **1008**. the first portion **1004** can be removed from the K wires **1300**. The second portion **1008** can be removed from the k-wires **1300**.
(115) As shown in FIG. **32**, the resection guide **1100** can be slid over the k-wires **1300**. The planar portion **1109** can be aligned with one or both of the medial cuneiform bone **104** and/or the metatarsal **108**. A resection tool **1400** can be inserted through the slot **1107** to form resection planes **104***a*, and/or **108***a* on the respective medial cuneiform bone **104** and metatarsal **108**. As described above, this can facilitate alignment of the medial cuneiform bone **104** and the metatarsal **108** in a corrected configuration **103**.
(116) As shown in FIG. **33**, a correction guide **1500** can be slid over the k-wires **1300**. The correction guide **1500** can be similar to the correction guide **500**. The correction guide **1500** can include a plurality of cannula extending along parallel axes. The k-wires **1300** can be received in the cannula of the collection guide **1500**. This can realign and adjust positions of the medial cuneiform bone **104**, metatarsal **108** and/or the proximal phalanx **112** to form the corrected configuration **103** of the patient's foot **100**.
(117) In the corrected configuration **103**, a fixing k-wire **1600** (or similar mechanism) can be inserted to fix the positions of the first metatarsal **108** and the medial cuneiform bone **104**. As shown in FIG. **34**, a bone plate assembly **1700**, similar to the bone plate assembly **700**, can be attached to the medial cuneiform bone **104** and the metatarsal **108** to maintain the relative positions of the two bones in the corrected configuration **103**.
(118) Certain Terminology
(119) Terms of orientation used herein, such as “top,” “bottom,” “proximal,” “distal,” “longitudinal,” “lateral,” and “end,” are used in the context of the illustrated example. However, the present disclosure should not be limited to the illustrated orientation. Indeed, other orientations are possible and are within the scope of this disclosure. Terms relating to circular shapes as used herein, such as diameter or radius, should be understood not to require perfect circular structures, but rather should be applied to any suitable structure with a cross-sectional region that can be measured from side-to-side. Terms relating to shapes generally, such as “circular,” “cylindrical,” “semi-circular,” or “semi-cylindrical” or any related or similar terms, are not required to conform strictly to the mathematical definitions of circles or cylinders or other structures, but can encompass structures that are reasonably close approximations.
(120) Conditional language, such as “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain examples include or do not include, certain features, elements, and/or steps. Thus, such conditional language is not generally intended to imply that features, elements, and/or steps are in any way required for one or more examples.

(121) Conjunctive language, such as the phrase “at least one of X, Y, and Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be either X, Y, or Z. Thus, such conjunctive language is not generally intended to imply that certain examples require the presence of at least one of X, at least one of Y, and at least one of Z.
(122) The terms “approximately,” “about,” and “substantially” as used herein represent an amount close to the stated amount that still performs a desired function or achieves a desired result. For example, in some examples, as the context may dictate, the terms “approximately,” “about,” and “substantially,” may refer to an amount that is within less than or equal to 10% of the stated amount. The term “generally” as used herein represents a value, amount, or characteristic that predominantly includes or tends toward a particular value, amount, or characteristic. As an example, in certain examples, as the context may dictate, the term “generally parallel” can refer to something that departs from exactly parallel by less than or equal to 20 degrees. All ranges are inclusive of endpoints.

SUMMARY

(123) Several illustrative examples of Lapidus procedure systems and methods have been disclosed. Although this disclosure has been described in terms of certain illustrative examples and uses, other examples and other uses, including examples and uses which do not provide all of the features and advantages set forth herein, are also within the scope of this disclosure. Components, elements, features, acts, or steps can be arranged or performed differently than described and components, elements, features, acts, or steps can be combined, merged, added, or left out in various examples. All possible combinations and subcombinations of elements and components described herein are intended to be included in this disclosure. No single feature or group of features is necessary or indispensable.
(124) Certain features that are described in this disclosure in the context of separate implementations can also be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation also can be implemented in multiple implementations separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations, one or more features from a claimed combination can in some cases be excised from the combination, and the combination may be claimed as a subcombination or variation of a subcombination.
(125) Any portion of any of the steps, processes, structures, and/or devices disclosed or illustrated in one example in this disclosure can be combined or used with (or instead of) any other portion of any of the steps, processes, structures, and/or devices disclosed or illustrated in a different example or flowchart. The examples described herein are not intended to be discrete and separate from each other. Combinations, variations, and some implementations of the disclosed features are within the scope of this disclosure.
(126) While operations may be depicted in the drawings or described in the specification in a particular order, such operations need not be performed in the particular order shown or in sequential order, or that all operations be performed, to achieve desirable results. Other operations that are not depicted or described can be incorporated in the example methods and processes. For example, one or more additional operations can be performed before, after, simultaneously, or between any of the described operations. Additionally, the operations may be rearranged or reordered in some implementations. Also, the separation of various components in the implementations described above should not be understood as requiring such separation in all implementations, and it should be understood that the described components and systems can generally be integrated together in a single product or packaged into multiple products. Additionally, some implementations are within the scope of this disclosure.
(127) Further, while illustrative examples have been described, any examples having equivalent elements, modifications, omissions, and/or combinations are also within the scope of this

disclosure. Moreover, although certain aspects, advantages, and novel features are described herein, not necessarily all such advantages may be achieved in accordance with any particular example. For example, some examples within the scope of this disclosure achieve one advantage, or a group of advantages, as taught herein without necessarily achieving other advantages taught or suggested herein. Further, some examples may achieve different advantages than those taught or suggested herein.

(128) Some examples have been described in connection with the accompanying drawings. The figures are drawn and/or shown to scale, but such scale should not be limiting, since dimensions and proportions other than what are shown are contemplated and are within the scope of the disclosed invention. Distances, angles, etc. are merely illustrative and do not necessarily bear an exact relationship to actual dimensions and layout of the devices illustrated. Components can be added, removed, and/or rearranged. Further, the disclosure herein of any particular feature, aspect, method, property, characteristic, quality, attribute, element, or the like in connection with various examples can be used in all other examples set forth herein. Additionally, any methods described herein may be practiced using any device suitable for performing the recited steps.

(129) For purposes of summarizing the disclosure, certain aspects, advantages and features of the inventions have been described herein. Not all, or any such advantages are necessarily achieved in accordance with any particular example of the inventions disclosed herein. No aspects of this disclosure are essential or indispensable. In many examples, the devices, systems, and methods may be configured differently than illustrated in the figures or description herein. For example, various functionalities provided by the illustrated modules can be combined, rearranged, added, or deleted. In some implementations, additional or different processors or modules may perform some or all of the functionalities described with reference to the examples described and illustrated in the figures. Many implementation variations are possible. Any of the features, structures, steps, or processes disclosed in this specification can be included in any example.

(130) In summary, various examples of Lapidus procedure systems and related methods have been disclosed. This disclosure extends beyond the specifically disclosed examples to other alternative examples and/or other uses of the examples, as well as to certain modifications and equivalents thereof. Moreover, this disclosure expressly contemplates that various features and aspects of the disclosed examples can be combined with, or substituted for, one another. Accordingly, the scope of this disclosure should not be limited by the particular disclosed examples described above, but should be determined only by a fair reading of the claims.

Claims

1. A kit for correcting alignment between a first bone and a second bone by fusing a joint between the first bone and the second bone, the kit comprising: a first guide comprising: a first end portion having a first cannula aligned along a first axis; a second end portion having a second cannula aligned along a second axis, the first axis being non-parallel with the second axis; wherein the first guide is configured such that inserting a first k-wire through the first cannula intersects the first bone and inserting a second k-wire through the second cannula intersects the second bone when the first and second bones are in a deformed configuration; a second guide comprising: a first end portion having a first cannula; and a second end portion having a second cannula, the first cannula being parallel with the second cannula, wherein the first cannulas of the first and second guides have a respective first same diameter, and the second cannulas of the first and second guides have a respective second same diameter; wherein the second guide is configured such that when the first k-wire is fixed within the first bone and the second k-wire is fixed within the second bone in the deformed configuration, sliding the second guide over first and second k-wires, with the first and second k-wires being received within the respective first and second cannula of the second guide, re-aligns the first and second bones into a corrected configuration, and wherein the first guide and

the second guide are separate structures.
2. The kit of claim 1, further comprising: a stabilizing wire configured to be inserted into the first and second bone so as to fix the first and second bones in the corrected configuration.
3. The kit of claim 1, further comprising: a bone plate, a first end configured to be attached with the first bone and a second end of the bone plate configured to be attached with the second bone such that the first and second bones are retained in the corrected configuration.
4. The kit of claim 3, further comprising a bone plate clip configured for insertion into the first and second bones in the corrected configuration.
5. The kit of claim 1, further comprising: a first resecting guide configured to align a resecting tool with a resection location on the first bone.
6. The kit of claim 5, wherein the first resecting guide includes first and second cannula configured to be advanced over the first k-wire and a third k-wire, the third k-wire being parallel with the first k-wire.
7. The kit of claim 6, further comprising a second resecting guide configured to align the resecting tool with a resection location on the second bone.
8. The kit of claim 1, wherein the first end portion of the first guide includes a third cannula aligned parallel with the first cannula, the third cannula configured to receive a third k-wire that intersects the first bone.
9. The kit of claim 8, wherein the second end portion of the first guide includes a fourth cannula aligned parallel with the second cannula, the fourth cannula configured to receive a fourth k-wire that intersects the second bone.
10. The kit of claim 9, wherein the second guide includes third and fourth cannula, wherein the second guide is configured to be slid over the first and second k-wires, and the third and fourth k-wires are configured to be slid over the respective third and fourth cannula of the second guide to re-align the first and second bones into the corrected configuration.
11. The kit of claim 1, wherein the first and second cannula of the second guide are spaced a distance such that the first and second K-wires are configured to translate the first bone relative to the second bone into abutment in the corrected configuration when the first and second K-wires are received in the first and second cannula, respectively.
12. The kit of claim 1, wherein the first and second portions of the first guide are coupled together by an attachment mechanism.
13. The kit of claim 1, wherein the first guide includes a removable tube coupled with the first cannula.
14. The kit of claim 1, wherein the first axis and the second axis define a correction factor of the first guide.
15. The kit of claim 1, wherein the first same diameter is equal to the second same diameter.
16. The kit of claim 1, wherein the first same diameter is different than the second same diameter.
17. The kit of claim 1, wherein the first and second cannulas of the first guide are aligned along a first major length of the first guide, and the first and second cannulas of the second guide are aligned along a second major length of the second guide.
18. The kit of claim 17, wherein the first and second cannulas of the first guide have different diameters.
19. The kit of claim 18, wherein the first and second cannulas of the second guide have different diameters.